

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 365 trials
Total training paths	1 · 365 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats+fft → feature dim 9504
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=292 · test=73

Augmentation (training only)

Enabled	no
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (Linear)	0.9966	0.0103	1.0000
Random Forest	1.0000	0.0000	1.0000
LDA	0.9966	0.0103	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.9897	0.0158	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9931	0.0138	1.0000
Logistic Regression	0.9828	0.0231	0.9863
SVM (RBF)	0.9862	0.0229	0.9726
KNN	0.9554	0.0410	0.9315

Per-class hold-out F1

Class	SVM (Linear)	Random Forest	LDA	Voting (soft)	Stacking	Logistic Regression	SVM (RBF)	KNN	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.929	13
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	0.957	0.960	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.923	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	0.957	0.957	0.800	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	0.960	0.923	1.000	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.957	12
macro avg	1.000	1.000	1.000	1.000	1.000	0.986	0.973	0.928	73

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000
LDA	0.8387
Voting Soft (SVM+RF+KNN+LogReg)	0.8387
Stacking (SVM+RF+KNN+LogReg)	0.8387
SVM (RBF)	0.8065
SVM (Linear)	0.8065
Logistic Regression	0.7742
KNN	0.7097

Per-class external accuracy

[illegible]